



Set Operations & Algebraic Laws

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SUBJECT

Mathematics Foundation T

SEMESTER

Semester 1

// Notes Summary

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title: "Set Operations Algebraic Laws" semester:
subject: "Mathematics Foundation to Computer Science
I" subjectSlug: "mathfoundation1" unit: "Unit I:
Set, Relation and Function" unitSlug:
"unit1setrelationfunction" order: description:
"Union, intersection, set difference, absolute
```

NOTES THAT STICK.

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Topics Covered

Intermediate

Difficulty Level



Why this topic exists

Set operations provide the mathematical foundation for manipulating data collections, evaluating Boolean logic expressions, and constructing database queries (such as SQL `UNION`, `INTERSECT`, and `EXCEPT`). Just as numerical arithmetic follows specific algebraic rules, set operations adhere to formal laws. Mastering these properties enables expression simplification, logical proof construction, and query optimization in computer science.

The Four Main Operations

✦ [DEFINITION] 1. Union ($A \cup B$)

Contains all elements belonging to A , B , or both.

Formal Notation: $A \cup B = \{x \mid x \in A \vee x \in B\}$

Example: If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \cup B = \{1, 2, 3, 4, 5\}$.

CS Parallel: Corresponds to the logical **OR** ($\vee$) operator and SQL `UNION` command.

✦ [DEFINITION] 2. Intersection ($A \cap B$)

Contains only the elements common to both A and B .

Formal Notation: $A \cap B = \{x \mid x \in A \wedge x \in B\}$

Example: If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \cap B = \{3\}$.

CS Parallel: Corresponds to the logical **AND** ($\wedge$) operator and SQL `INTERSECT` command.

✦ [DEFINITION] 3. Difference ($A - B$ or $A \setminus B$)

Contains elements that belong to A but do not belong to B .

Formal Notation: $A - B = \{x \mid x \in A \wedge x \notin B\}$

Example: If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then: $A - B = \{1, 2\}$ - $B - A = \{4, 5\}$

CS Parallel: Corresponds to SQL `EXCEPT` or `MINUS`. Note that set difference is non-commutative ($A - B \neq B - A$).

✦ [DEFINITION] 4. Complement (A' or A^c)

Contains all elements in the Universal Set (U) that do not belong to A .

Formal Notation: $A' = U - A = \{x \mid x \in U \wedge x \notin A\}$

Example: If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2, 3\}$, then $A' = \{4, 5\}$.

CS Parallel: Corresponds to the logical **NOT** ($\neg$) operator.

Properties of Set Operations

These algebraic laws govern set operations, allowing symbolic simplification of set expressions without enumerating individual elements.

Commutative Laws Order does not alter the outcome of a union or intersection:

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative Laws Grouping does not alter the outcome when executing sequential unions or intersections:

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Distributive Laws Union distributes over intersection, and intersection distributes over union:



$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Identity Laws Union with the empty set ($\emptyset$) or intersection with the universal set (U) preserves the original set:

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Complement Laws Combining a set with its complement yields boundary sets (the full universe or an empty set):

$$A \cup A' = U$$

$$A \cap A' = \emptyset$$

Summary Table of Properties

Law Name	Union Formulation	Intersection Formulation
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A$	$A \cap U = A$
Complement	$A \cup A' = U$	$A \cap A' = \emptyset$